

COMP9334 Revision Questions for Week 4B

Question 1

The Python function *sim_mm1_func()* (in the file *sim_mm1_lib.py*) simulates an M/M/1 queue with arrival rate λ and service rate μ over a time period of T . It returns the average response time for the given λ , μ and T .

We would like to investigate the effect of the length of simulation T on the simulation. Let us fix $\lambda = 0.7$ and $\mu = 1$. For each of the following values of T : 1000, 5000, 10000 and 50000, perform the simulation 20 times (with a different set of random numbers) and record the value of the average mean response time.

Answer the following:

1. What is the mean response time according to the M/M/1 result?
2. For each value of T used, compute the mean and standard deviation over 20 experiments.
3. How does the standard deviation vary with T ?

Note the Python file *sim_mm1_lib.py* contains the Python function *sim_mm1_func()*. If you call the function *sim_mm1_func(0.7,1,1000)*, it will simulate an M/M/1 queue with $\lambda = 0.7$, $\mu = 1$ and $T = 1000$. The easiest is to import the file and then call the function.

Question 2

Write a simulation program (in whatever language you prefer) to simulate an M/M/2 queue. You should be able to control the arrival rate λ , service rate μ and the length of simulation T .

Use your M/M/2 and M/M/1 simulation program to compare:

1. The mean response time of an M/M/1 queue with $\lambda = 0.9$ and $\mu = 1$.
2. The mean response time of an M/M/2 queue with $\lambda = 0.9$ and for each server, $\mu = 0.5$.

Simulate each of the above configurations 10 times and record the mean response time in the simulation. You may use a simulation time of $T = 1000$.

You have learnt in Week 3 that the the first system should have a smaller mean response time. Did your simulation results also suggest a smaller mean response time for the first system?